EEE373L

Embedded System Design Laboratory

Lab Report

Section:02 Group No:02

Experiment no:02

Name of the experiment: **Fundamentals of Embedded C**

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Objective:

The objective of this experiment is to illustrate the foundations of programming in the Embedded C language.

Codes:

Part A:

```
#include <mega32.h> // library for ATmega32 microcontroller

#include <alcd.h>   // library for LCD functions


unsigned char globey; // A global unsigned character variable 'globey' declared


void function_z (void) // A function_z is defined

{

    unsigned int tween; //A local unsigned integer variable 'tween' declared

    tween = 12; // Assign the value 12 to 'tween'

    globey = 47; // Assign the value 47 to the global variable 'globey'

    main_loc = 12; // ‘main_loc' is assigned as 12

}


void main(void) // The main function is defined

{

    unsigned char main_loc; //A local unsigned character variable 'main_loc' 
declared

    lcd_init(16); // Initializing the LCD with 16 columns

    globey = 34; // The value 34 is assigned to the global variable 'globey'

    tween = 12; // 12 is assigned to 'tween'

    while (1) // Generates infinite loop

    {

    }

}
```

Discussion:

There are two errors generated because the variable `main_loc` is used in `function_z`, but it is not declared within the scope of the function. This results in an error because the function has no access to the `main_loc` declared in the `main` function. Again, the variable `tween` is used in the `main` function without being declared within its scope. This results in an error because `tween` is not accessible outside of `function_z`, where it is declared locally.

Part B:

Code:

```
#include <mega32.h>   // library for ATmega32 microcontroller

#include <alcd.h>     // library for LCD functions

#include <delay.h>    // library for delay functions

#include <stdlib.h>   // standard library for functions like itoa

#include <string.h>   // string library for functions like strcpy and strcat


char k; // A global character variable 'k' is declared


void display(char var, char disp1[16]) // Function to display a variable on the
LCD

{

    char disp2[16];       // Declares a local character array 'disp2' of size
16

    itoa(var,disp2);      // Converts the integer 'var' to a string and store
it in 'disp2'

    strcat(disp1,disp2);  // Concatenates 'disp1' and 'disp2'

    lcd_gotoxy(0,1);      // Sets the cursor to the first position of the
second row on the LCD

    lcd_puts(disp1);      // Displays the string 'disp1' on the LCD

}


void main(void) // Declares main function
```

```
{

    char i, disp1[16]; // Declares local variables 'i' and 'disp1' (array of
size 16)

    lcd_init(16);      // Initializes the LCD with 16 columns


    while (1) // Generates infinite loop

    {

        i = 1; // Initializes 'i' with 1

        k = 4 * i++; // Assigns 'k' the value of 4 multiplied by 'i', then
increment 'i'

        lcd_clear(); // Clears the LCD

        lcd_gotoxy(0,0); // Sets the cursor to the first position of the first
row on the LCD

        lcd_putsf("Post-incr "); // Displays the string "Post-incr" on the LCD

        strcpy(disp1, "k = "); // Copies the string "k = " into 'disp1'

        display(k, disp1); // Calls the display function to show 'k' on the LCD

        delay_ms(1000); // Delays for 1000 milliseconds

        i = 1; // Initializes 'i' with 1

        k = 4 * ++i; // Increments 'i' first, then assigns 'k' the value of 4
multiplied by 'i'

        lcd_gotoxy(0,0); // Sets the cursor to the first position of the first
row on the LCD

        lcd_putsf("Pre-incr"); // Displays the string "Pre-incr" on the LCD

        strcpy(disp1, "k = "); // Copies the string "k = " into 'disp1'

        display(k, disp1); // Calls the display function to show 'k' on the LCD

        delay_ms(1000); // Delays for 1000 milliseconds


        // i += 2; //increments 'i' by 2

        k /= i; // Divides 'k' by 'i' and assign the result to 'k' (k = 4)

        k |= 5; // Performs bitwise OR on 'k' with 5 and assign the result to
'k' (k = 5)
```

```
        k &= 3; // Performs bitwise AND on 'k' with 3 and assign the result to
'k' (k = 1)

        k ^= 7; // Performs bitwise XOR on 'k' with 7 and assign the result to
'k' (k = 6)

        lcd_clear(); // Clears the LCD

        lcd_gotoxy(0,0); // Sets the cursor to the first position of the first
row on the LCD

        lcd_putsf("Compound  "); // Displays the string "Compound" on the LCD

        strcpy(disp1, "k = "); // Copies the string "k = " into 'disp1'

        display(k, disp1); // Calls the display function to show 'k' on the LCD

        delay_ms(1000); // Delays for 1000 milliseconds

    }

}
```

Output:

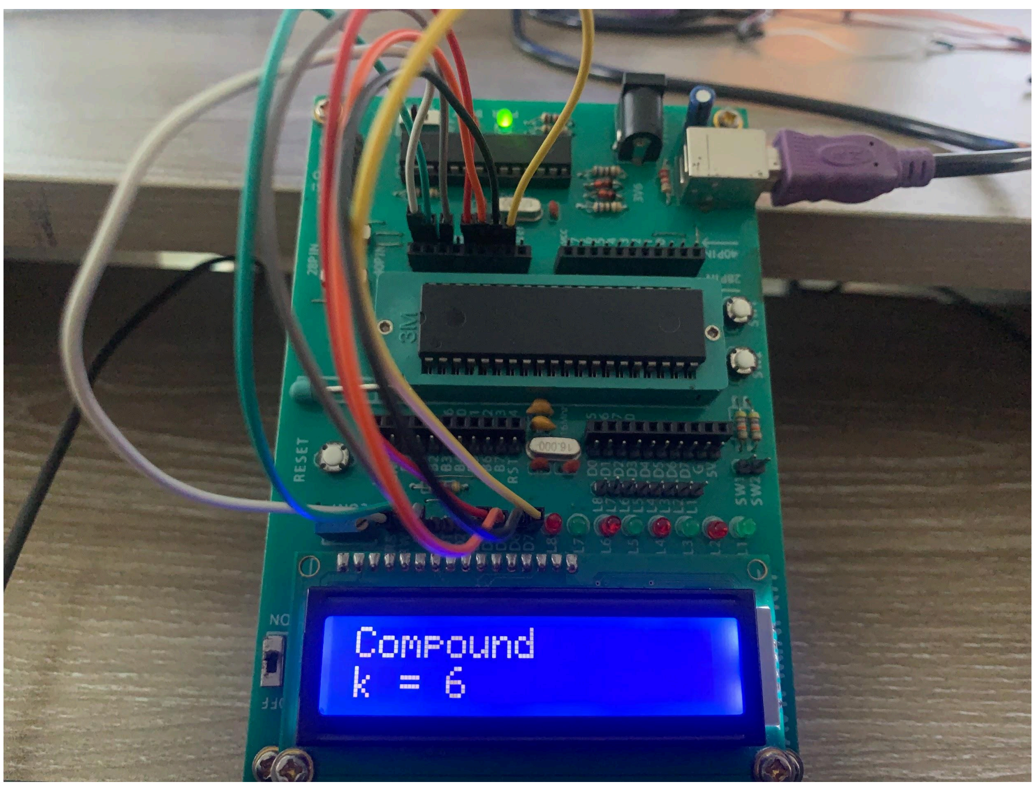


Figure-1: Output using Assignment statement

Part C:

Code:

```
#include <mega32.h>   // library for ATmega32 microcontroller

#include <alcd.h>     // library for LCD functions

#include <stdlib.h>   // standard library for functions like itoa

#include <string.h>   // string library for functions like strcpy and strcat

#include <delay.h>    // library for delay functions


void display(char var, char disp1[16]) // Function to display a variable on the
LCD

{

    char disp2[16];       // Declares a local array 'disp2' of size 16

    itoa(var, disp2);     // Converts the integer 'var' to a string and store
it in 'disp2'

    strcat(disp1, disp2); // Concatenates 'disp1' and 'disp2'

    lcd_gotoxy(0, 1);     // Sets the cursor to the first position of the
second row on the LCD

    lcd_puts(disp1);      // Displays the string 'disp1' on the LCD

}


void main(void) // Declares Main function

{

    char i, k, disp1[16]; // Declares local variables 'i', 'k', and 'disp1'
(array of size 16)

    lcd_init(16);         // Initializes the LCD with 16 columns


    while (1) // Generates infinite loop

    {

        i = 0; // Initializes 'i' with 0

        k = 0; // Initializes 'k' with 0
```

```
        // While loop upto 20
        while (i < 20) // Loop until 'i' is less than 20
        {
            k += 3; // Increments 'k' by 3
            i++;    // Increments 'i' by 1
        }

        lcd_gotoxy(0, 0);  // Sets the cursor to the first position of the 
first row on the LCD
        lcd_puts("while loop..."); // Displays the string "while loop..." on 
the LCD
        strcpy(disp1, "k = "); // Copies the string "k = " into 'disp1'
        display(k, disp1); // Calls the display function to show 'k' on the LCD
        delay_ms(1000); // Delays for 1000 milliseconds

        i = 0; // Reinitializes 'i' with 0
        k = 0; // Reinitializes 'k' with 0

        // Do-while loop starts
        do // Execute the loop body at least once
        {
            k += 3; // Increments 'k' by 3
            i++;    // Increments 'i' by 1
        }
        while (i < 20); // Continues looping until 'i' is less than 20

        lcd_gotoxy(0, 0); // Sets the cursor to the first position of the first 
row on the LCD
```

```
        lcd_puts("do while ..."); // Displays the string "do while ..." on the LCD

        strcpy(disp1, "k = "); // Copies the string "k = " into 'disp1'

        display(k, disp1); // Calls the display function to show 'k' on the LCD

        delay_ms(1000); // Delays for 1000 milliseconds

    }

}
```

Output:

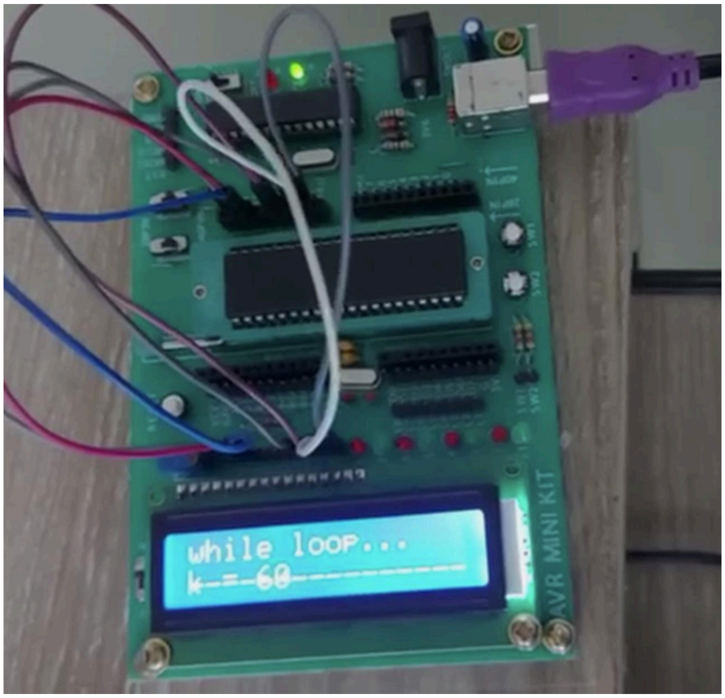


Figure-2,3: Output using While and Do While loop

Part D:

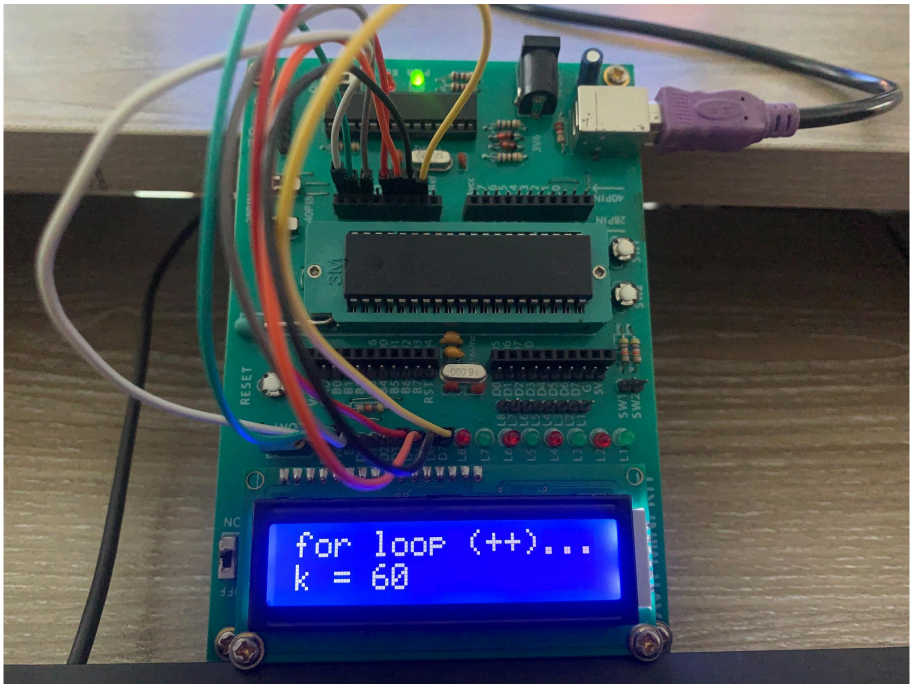
Code:

```
#include <mega32.h>   // library for ATmega32 microcontroller

#include <alcd.h>     // library for LCD functions

#include <stdlib.h>   // standard library for functions like itoa

#include <string.h>   // string library for functions like strcpy and strcat

#include <delay.h>    // library for delay functions


// Function to display a variable on the LCD

void display(char var, char disp1[16])

{

    char disp2[16];       // Declares a local array 'disp2' of size 16

    itoa(var, disp2);     // Converts the integer 'var' to a string and store
it in 'disp2'

    strcat(disp1, disp2); // Concatenates 'disp1' and 'disp2'
```

```
    lcd_gotoxy(0, 1);      // Sets the cursor to the first position of the
second row on the LCD

    lcd_puts(disp1);       // Displays the string 'disp1' on the LCD

}


void main(void) // Declares Main function

{

    char i, k, disp1[16]; // Declares local variables 'i', 'k', and 'disp1'
(array of size 16)

    lcd_init(16);          // Initializes the LCD with 16 columns


    while (1) // Generates infinite loop

    {

        k = 0; // Initializes 'k' with 0


        // For loop with incrementing 'i'

        for (i = 0; i < 20; i++) // Loop from 0 to 19

        {

            k += 3; // Increments 'k' by 3 in each iteration

        }


        lcd_clear();               // Clears the LCD

        lcd_gotoxy(0, 0);          // Sets the cursor to the first position of
the first row on the LCD

        lcd_puts("for loop (++)..."); // Displays the string "for loop (++)..."
on the LCD

        strcpy(disp1, "k = ");     // Copies the string "k = " into 'disp1'

        display(k, disp1);         // Calls the display function to show 'k' on
the LCD

        delay_ms(1000);            // Delays for 1000 milliseconds
```

```
        k = 60; // Initializes 'k' with 60

        // For loop with decrementing 'i'
        for (i = 20; i > 0; i--) // Loop from 20 to 1
        {
            k -= 3; // Decrements 'k' by 3 in each iteration
        }

        lcd_clear();                // Clears the LCD
        lcd_gotoxy(0, 0);           // Sets the cursor to the first position of
the first row on the LCD
        lcd_puts("for loop (--)..."); // Displays the string "for loop (--)..."
on the LCD
        strcpy(disp1, "k = ");      // Copies the string "k = " into 'disp1'
        display(k, disp1);          // Calls the display function to show 'k' on
the LCD
        delay_ms(1000);             // Delays for 1000 milliseconds
    }
}
```

Output:



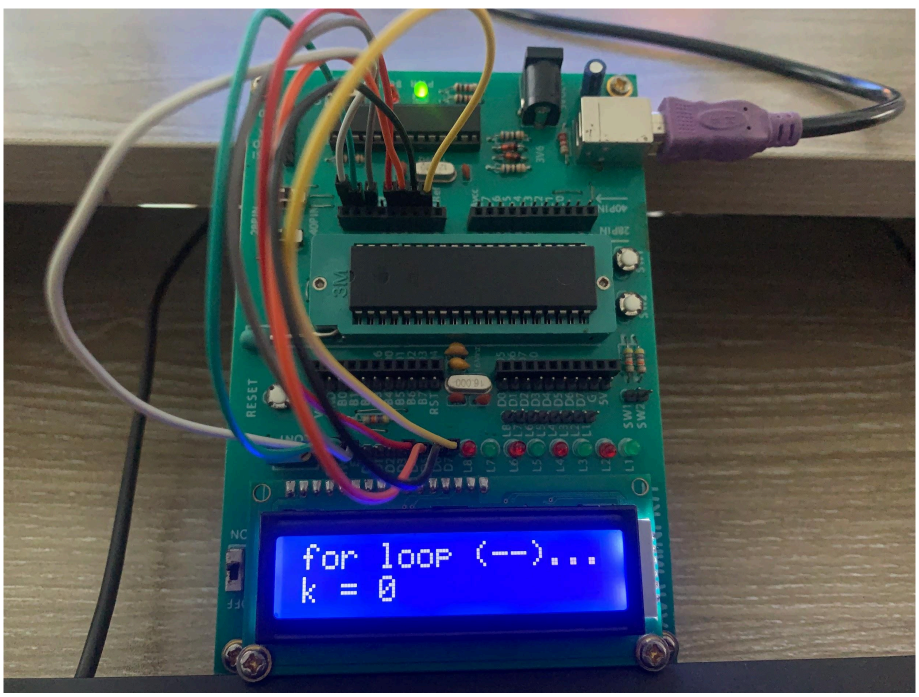


Figure-4,5: Output using For loop

<u>Part E:</u>

Code:

```
#include <mega32.h>    // library for ATmega32 microcontroller

#include <alcd.h>      // library for LCD functions
```

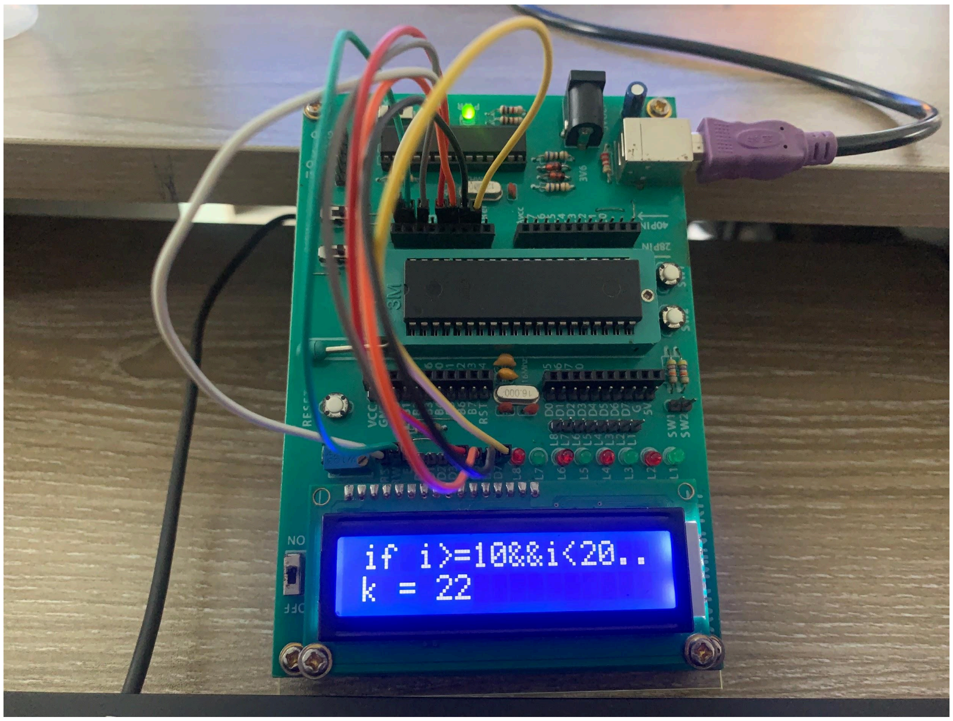
```
#include <stdlib.h>   // standard library for functions like itoa

#include <string.h>   // string library for functions like strcpy and strcat

#include <delay.h>    // library for delay functions


// Function to display a variable on the LCD

void display(char var, char disp1[16])

{

    char disp2[16];       // Declares a local array 'disp2' of size 16

    itoa(var, disp2);     // Converts the integer 'var' to a string and store
it in 'disp2'

    strcat(disp1, disp2); // Concatenates 'disp1' and 'disp2'

    lcd_gotoxy(0, 1);     // Sets the cursor to the first position of the
second row on the LCD

    lcd_puts(disp1);      // Displays the string 'disp1' on the LCD

}


void main(void) // Declares main function

{

    char i, k, disp1[16]; // Declares local variables 'i', 'k', and 'disp1'
(array of size 16)

    lcd_init(16);         // Initializes the LCD with 16 columns


    while (1) //Generates infinite loop

    {

        // if-else if-else statement

        i = 15; // Initializes 'i' with 15

        k = 7;  // Initializes 'k' with 7

        lcd_clear();          // Clears the LCD

        lcd_gotoxy(0, 0);     // Sets the cursor to the first position of the
first row on the LCD
```

```
        if (i < 10) // If 'i' is less than 10

        {

            lcd_puts("if i<10 ........"); // Displays the string "if i<10
........" on the LCD

            k += 10; // Increments 'k' by 10

        }

        else if (i >= 10 && i < 20) // If 'i' is between 10 and 19 inclusive

        {

            lcd_puts("if i>=10&&i<20.."); // Displays the string "if
i>=10&&i<20.." on the LCD

            k += 15; // Increments 'k' by 15

        }

        else // If 'i' is 20 or more

        {

            lcd_puts("if i>=20 ........"); // Displays the string "if i>=20
........" on the LCD

            k += 20; // Increments 'k' by 20

        }


        strcpy(disp1, "k = "); // Copies the string "k = " into 'disp1'

        display(k, disp1);     // Calls the display function to show 'k' on the
LCD

        delay_ms(1000);        // Delays for 1000 milliseconds


        // Switch statement

        i = 1; // Initializes 'i' with 1

        k = 7; // Initializes 'k' with 7

        lcd_clear();           // Clears the LCD
```

```
        lcd_gotoxy(0, 0);       // Sets the cursor to the first position of the
first row on the LCD


        switch (i) // Switches statement based on the value of 'i'

        {

            case 1: // If 'i' is 1

                lcd_puts("switch i=1 ....."); // Displays the string "switch
i=1 ....." on the LCD

                k += 10; // Increments 'k' by 10

                break;   // Exits the switch statement

            case 2: // If 'i' is 2

                lcd_puts("switch i=2......."); // Displays the string "switch
i=2......." on the LCD

                k += 15; // Increments 'k' by 15

                break;   // Exits the switch statement

            case 3: // If 'i' is 3

                lcd_puts("switch i=3......."); // Displays the string "switch
i=3......." on the LCD

                k += 20; // Increments 'k' by 20

                break;   // Exits the switch statement

            default: // If 'i' is any other value

                lcd_puts("switch default...."); // Displays the string "switch
default...." on the LCD

                k += 0; // Do not change the value of 'k'

                break;  // Exits the switch statement

        }


        strcpy(disp1, "k = "); // Copies the string "k = " into 'disp1'

        display(k, disp1);     // Calls the display function to show 'k' on the
LCD

        delay_ms(1000);        // Delays for 1000 milliseconds
```

```
        }
}
```

Output:



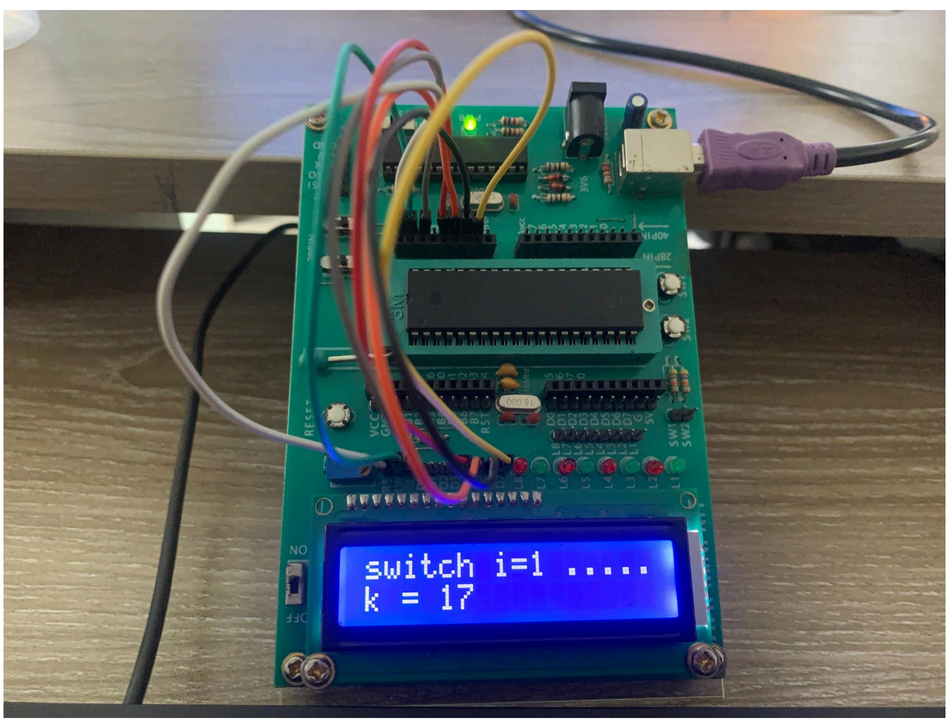


Figure-6,7: Output using if else and switch statement